

Can Middle-School Science Textbooks Help Students Learn Important Ideas? Findings from Project 2061's Curriculum Evaluation Study: Life Science

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Abstract: The transfer of matter and energy from one organism to another and between organisms and their physical setting is a fundamental concept in life science. Not surprisingly, this concept is common to the *Benchmarks for Science Literacy* (American Association for the Advancement of Science, 1993), the *National Science Education Standards* (National Research Council, 1996), and most state frameworks and likely to appear in any middle-school science curriculum material. Nonetheless, while topics such as photosynthesis and cellular respiration have been taught for many years, research on student learning indicates that students have difficulties learning these ideas. In this study, nine middle-school curriculum materials—both widely used and newly developed—were examined in detail for their support of student learning ideas concerning matter and energy transformations in ecosystems specified in the national standards documents. The analysis procedure used in this study was previously developed and field tested by Project 2061 of the AAAS on a variety of curriculum materials. According to our findings, currently available curriculum materials provide little support for the attainment of the key ideas chosen for this study. In general, these materials do not take into account students' prior knowledge, lack representations to clarify abstract ideas, and are deficient in phenomena that can be explained by the key ideas and hence can make them plausible. This article concludes with a discussion of the implications of this study to curriculum development, teaching, and science education research based on shortcomings in today's curricula.

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Ideas about how matter cycles and energy flows from one living thing to another and between organisms and their environment are key scientific concepts that bring together insights from the physical and biological sciences. Transformations of matter and energy can be found at many levels of biological organization, from molecules to ecosystems (AAAS, 1993), and are relevant to

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everyday life and current societal issues (Carlsson, 2002a). Therefore, it is not surprising that Project 2061's *Benchmarks for Science Literacy* (AAAS, 1993), the *National Science Education Standards* (NSES) of the NRC (1996), and most state frameworks recommend that to become science-literate adults, middle- and high-school students should know these core concepts.

Topics related to the flow of matter and energy in ecosystems, such as the food making process, the release of energy from food, and the flow of matter and energy in the food web, have been taught for many years and are likely to appear in all middle and high-school curriculum materials. Nonetheless, research on student learning indicates that even after relevant instruction, students have difficulties understanding ideas about food, plant and animal nutrition, and matter cycling (Anderson, Sheldon, & Dubay, 1990; Bell & Brook, 1984; Leach, Driver, Scott, & Wood-Robinson, 1996a,b; Roth & Anderson, 1987; Smith & Anderson, 1986; Stavy, Eisen, & Yaakobi, 1987; Wandersee, 1985; summaries of relevant research can be found in AAAS, 1993; Carlsson, 2002a; Driver, Squires, Rushworth, & Wood-Robinson, 1994). For example, students view matter as being created or destroyed rather than as being transformed (Smith & Anderson, 1986). Students who do see matter as being transformed view it as being transformed into energy rather than into simpler substances, and have difficulties appreciating the distinction between matter and energy in the context of ecosystems (Leach et al., 1996a,b). In terms of matter transformation, students often think that organisms and materials in the environment are very different types of matter and are not transformable into each other or that food is a requirement for growth rather than a source of matter for growth (Smith & Anderson, 1986). Furthermore, middle and high-school students have little understanding about food being transformed and constituting the organism's body (Leach et al., 1996a,b; Schneps & Sadler, 1988; Smith & Anderson, 1986). When students of all ages were asked to explain where the mass of a tree comes from, most said water and soil, ignoring the contribution of carbon dioxide. When told that plants make food from water and air and that some of this food is transformed into the plant body, most students were reluctant to believe it (Anderson et al., 1990; Barker & Carr, 1989; Roth & Anderson, 1987; Schneps & Sadler, 1988; Wandersee, 1985).

While textbooks are not singly to blame for all the problems in student learning, they largely determine *what* topics and ideas are taught in the classrooms and *how* these topics are taught (Association for Supervision and Curriculum Development, 1997; Tyson, 1997). A study conducted 20 years ago found that 90% of all science teachers use a textbook 95% of the time (Harmes & Yager, 1981, as cited in Renner, Abraham, Grzybowski, & Marek, 1990). More recent studies indicated that many teachers rely on curriculum materials to provide them with some or all the content or the pedagogical content knowledge (Ball & Feiman-Nemser, 1988; National Educational Goals Panel, 1994). Reliance on curriculum materials is more apparent when teachers are teaching outside their own area of expertise, which is typically the case at the middle-school level when science teachers often have majored in education rather than in science. Poor curriculum materials can deprive both students and teachers of ways that allow them to understand and implement effective teaching practices (Abraham, Grzybowski, Renner, & Marek, 1992). Nonetheless, when used properly, good curriculum materials can be a powerful catalyst for improving teaching and learning (Ball & Cohen, 1996; Schmidt, McKnight, & Raizen, 1997). Indeed, some studies have suggested that textbooks that use effective teaching strategies improve student learning and provide good models for teaching (e.g., Bishop & Anderson, 1990; Lee, Eichinger, Anderson, Berkheimer, & Blakeslee, 1993). While better curriculum materials alone are unlikely to improve student learning, we think that high-quality curriculum materials can positively influence student learning directly and through their influence on teachers. For these reasons, valid identification of curriculum materials that actually support learning of worthwhile ideas and help teachers build their own content and pedagogical knowledge is essential.

For over 4 years, with input from hundreds of K–12 teachers, teacher educators, materials developers, scientists, and cognitive researchers, Project 2061 has developed and field tested a valid and reliable procedure for evaluating the credibility of science curriculum materials (Kesidou & Roseman, 2002; Kulm & Grier, 1998; Roseman, Kesidou, & Stern, 1996). The central premise in Project 2061's evaluation tool is that curriculum materials are to be judged primarily in terms of their likely contribution to the attainment of important and agreed-upon, specific learning goals such as benchmarks and national standards. While it is certainly important that materials are scientifically accurate, age appropriate, and motivating, if they do not also contribute significantly to students' learning important ideas and skills, they are inadequate for use. Hence, curriculum materials first are examined to determine whether they focus on ideas included in benchmarks and standards, and if so, whether instructional strategies in materials are consistent with what is currently known about how students learn these ideas. The criteria for making the judgments about instructional strategies were derived from published research on learning and effective teaching from the past two decades. This includes the importance of building on prior knowledge, the use of relevant phenomena for making scientific ideas plausible and representations for making abstract ideas intelligible, the importance of guiding students' interpretation of their learning experiences, and the need to facilitate the transfer of knowledge (e.g., Lee et al., 1993; NRC, 2000; Posner, Strike, Hewson, & Gertzog, 1982; Smith, Blakeslee, & Anderson, 1993). The criteria examine student texts and teachers' guides and take account of first-hand experiences as well as of text. They are consistent with equity concerns and take into consideration both individual and social aspects of learning (Kesidou & Roseman, 2002).

Based on this procedure, Project 2061 has produced a database of analytical reports on middle-school curriculum materials in science (Kesidou & Roseman, 2002; full reports are available on Project 2061's Web site: <http://www.project2061.org/tools/textbook/mgsci/INDEX.HTM>). Educators and scientists, trained in the use of the Project 2061 procedure, examined nine curriculum materials—both widely used and newly developed—with respect to their treatment of fundamental topics from life, physical, and earth science. These topics—the kinetic molecular theory (physical science), flow of matter and energy in ecosystems (life science), and processes that shape the Earth (earth science)—serve as the basis for learning more complex ideas in middle and high school. They are included in *Benchmarks for Science Literacy* (AAAS, 1993), the NSES (NRC, 1996), and the majority of state frameworks (Kesidou & Roseman, 2003) and are treated by nearly all middle-school curriculum materials analyzed in our study. The sampling of three topics, rather than analyzing alignment of curricula to all middle-school benchmarks or fundamental concepts, had been done for pragmatic reasons given the rigor and time required for evaluating instructional quality of curriculum materials using Project 2061's analysis tool (Kesidou & Roseman, 2002, 2003; Stern, 2003; Stern & Ahlgren, 2002).

This article reports on how well ideas concerning matter cycling and energy flow in ecosystems are treated in the textbook series examined and describes typical findings to illustrate strengths and weaknesses in curriculum materials. The article concludes with a discussion of the implications of these findings for student learning and materials development.¹

Design and Procedures

Selection of Curriculum Materials

Nine comprehensive middle-school (spanning Grades 6–8) curriculum materials were examined in this study. They include both newly developed, commercially available materials

funded by the National Science Foundation and textbooks that are widely used or considered for use by selection committees. Data about use of curriculum materials in school districts and states were assembled by the 1998 Education Market Research report and surveys distributed by Project 2061 at the National Science Teachers Association conventions (see Table 1 for a complete list of the materials).

Selection and Clarification of Key Ideas in Life Science

As mentioned earlier, the resources required for evaluating curriculum materials using Project 2061's procedure make it impractical to use on every topic included in a yearlong or multiyear curriculum. Therefore, ideas from three topic areas advocated by *Benchmarks for Science Literacy* (AAAS, 1993), the NSES (NRC, 1996), and most state frameworks were used as the basis for the analysis (Kesidou & Roseman, 2003). In life science, the topic selected—flow of matter and energy in ecosystems—is an example of the core life science content likely to appear in any middle-school material and is familiar to teachers, policy makers, and parents. Furthermore, since much of the knowledge needed for effective teaching is specific to a particular idea (Shulman, 1986), and considerable cognitive research has been conducted about learning ideas related to matter and energy transformations and the kinetic molecular theory (on which it depends), research findings could be used to inform both curriculum development and curriculum evaluation using Project 2061's procedure.

Specific life science ideas within the topic “flow of matter and energy” were drawn from statements in both *Benchmarks for Science Literacy* (AAAS, 1993, chap. 5, section E) and the NSES (NRC, 1996, Life Science Content Standard C). These include ideas about matter transformations in living things (e.g., the idea that plants make sugars from carbon dioxide and water) as well as energy transformations in living things (e.g., the idea that plants use the energy from light to make “energy-rich” sugars). The complete list of key life science ideas is included next.

Life Science Ideas: Flow of Matter and Energy in Ecosystems

Idea a: Food (e.g., sugars) serves as fuel and building material for all organisms.

Idea b: Plants make their own food whereas animals obtain food by eating other organisms.

Matter is transformed in living systems (Ideas c1–c4):

Idea c1: Plants make sugars from carbon dioxide in the air and water.

Idea c2: Plants break down the sugars they have synthesized back into simpler substances: carbon dioxide and water; assemble sugars into the plants' body structures (including some energy stores).

Idea c3: Other organisms break down the stored sugars or the body structures of the plants they eat (or in the animals they eat) into simpler substances; reassemble them into their own body structures (including some energy stores).

Idea c4: Decomposers transform dead organisms into simpler substances, which other organisms can reuse.

Energy is transformed in living systems (Ideas d1–d3):

Idea d1: Plants use the energy from light to make “energy-rich” sugars.

Idea d2: Plants get energy by breaking down the sugars, releasing some of the energy as heat.

Table 1

Instructional scores in life science by instructional category or criterion (textbooks in alphabetical order)

Instructional Criterion	Textbook Series							
	<i>Glencoe: Life Science</i> (Glencoe/McGraw-Hill, 1997)	<i>Macmillan Science Series</i> (Macmillan McGraw-Hill, 1995)	<i>Prentice Hall: Exploring Life Science</i> (Prentice Hall, 1997)	<i>PRIME Science</i> (Kendall/Hunt, 1997)	<i>Science 2000</i> (Decision Development Corporation, 1995)	<i>Science Insights</i> (Addison Wesley, 1997)	<i>Science Interactions</i> (Glencoe/McGraw-Hill, 1997)	<i>Science Plus: Technology & Society</i> (Holt, Rinehart and Winston, 1997)
I. Providing a Sense of Purpose								
Conveying unit purpose	1	1.5	1.5	1	2.5	0.5	0.5	1.5
Conveying lesson purpose	1	1.5	0.5	1	1.5	0.5	1	2
Justifying activity sequence	1	1	1	1	2	1	1	2
II. Taking Account of Student Ideas								
Attending to prerequisite knowledge	0.5	0	0	1	1	0.5	1	1
Alerting teacher to commonly held ideas	0.5	1.5	0	0	0	0	0	0
Assisting teacher in identifying student ideas	1	1.5	0	0.5	2	0	1	2
Addressing commonly held ideas	0	1	0	0	0	0	0	1
III. Engaging Students with Relevant Phenomena								
Providing variety of phenomena	0.5	0.5	0.5	0.5	1	0	0.5	1
Providing vivid experiences	1	0.5	0.5	1	1.5	0	1	1
IV. Developing and Using Scientific Ideas								
Introducing terms meaningfully	0.5	1	0.5	1.5	1.5	1	0.5	2.5
Representing ideas effectively	0	1	0.5	1	1.5	0	1	0.5
Demonstrating use of knowledge	0	0	0	0.5	0	0	0	0
Providing practice	1	0.5	0.5	0.5	1	0.5	0.5	1
V. Promoting Student Thinking								
Encouraging students to explain their ideas	1	1	0	0.5	1.5	0	0.5	1
Guiding student interpretation and reasoning	0.5	1.5	0	1	1	0.5	0	0.5
Encouraging students to think about what they have learned	1	0	0	0	0.5	0	0.5	1
VI. Assessing Progress								
Aligning to goals	0.5	0.5	0.5	0.5	1	0.5	0.5	1.5
Testing for understanding	0	0	0.5	0	1	1	0.5	1
Informing instruction	0.25	0.5	0.25	0	0.5	0.5–1	0.5	0.5

0.1 = poor; 1.5 = fair; 2 = satisfactory; 2.5 = very good; 3 = excellent.

- Idea d3: Other organisms get energy to grow and function by breaking down the consumed body structures to sugars and then breaking down the sugars, releasing some of the energy into the environment as heat.
- Idea e: Matter and energy are transferred from one organism to another repeatedly and between organisms and their physical environment.

To obtain consistent analysis reports from independent teams and to ascertain valid interpretation of benchmarks and national standards, reviewers need to start with a precise understanding of their meaning. However, based on Project 2061's considerable experience, it was found that even specific and carefully crafted learning goals need to be clarified to ensure that reviewers neither ignore intended meanings nor read unintended meanings into the statements (Roseman, 1997; Stern & Roseman, 2001). For example, one of the learning goals that served as the basis of the analysis was "Food provides molecules that serve as fuel and building material for all organisms" (Idea a). Based on relevant commentary in *Benchmarks for Science Literacy* (AAAS, 1993) and relevant research on students' commonly held ideas and learning difficulties (described in the previous section), reviewers reached a common understanding of this particular learning goal. This idea is fundamental for understanding of the other key life science ideas, especially because the everyday meaning of the term "food" is inconsistent with its biological meaning. This idea goes beyond the idea that organisms *need* food to grow or for energy and addresses the more sophisticated idea that food is the *source* of both the building materials that make it possible for organisms to increase in mass and the fuel that provides the energy needed to carry out life functions (e.g., add new cells or convert inputs to outputs). During training, it was agreed that curriculum materials need to make the generalization (in text and in concrete examples) that *all* organisms—not only humans—get their building materials and energy from food; however, partial credit would be given to curriculum materials that treat either the matter or the energy side of the story. In a similar way, reviewers studied each key idea and reached a common understanding. Subsequent analyses of curriculum materials were based on this understanding (Kesidou & Roseman, 2003; Stern, 2003). The method for clarifying the precise meaning of each key idea is described in detail in Roseman (1997) and in Stern and Roseman (2001). Clarifications of all life science ideas that served as the basis of analysis can be found at Project 2061's Web site: www.project2061.org/tools/textbook/mgsci/ideas.htm.

Analytical Criteria

In essence, the Project 2061 evaluation procedure examines how well a material's content aligns with each key idea selected and how well the instructional strategies in the student text and the teacher's guide can support students' learning of this content.

Content Alignment. The procedure carefully examines materials' alignment to the specific key ideas selected, not just whether the material includes similar topic headings. At the topic-heading level, any science textbook appears to be aligned with standards. Nearly all middle-school textbooks use labels such as "Photosynthesis," "Cellular Respiration," or "Flow of Energy Through Ecosystems." But including the headings does not mean that textbooks necessarily address the specific ideas within those topics on which benchmarks and standards focus. For each curriculum material, reviewers examined the student text and the teacher's guide and referenced ancillary materials to identify text segments, activities, discussion questions, teacher notes, assessment, and any other materials (print or electronic) that addressed one or more of the key ideas. Each of these chunks was examined for content alignment.

To be considered “aligned” to a key idea, content in materials needs to address the *substance* rather than just the general topic of the learning goal. For example, the idea that “Plants use the energy from light to make ‘energy-rich’ sugars” (Idea d1) makes explicit the transformation of light energy into chemical energy (or “energy-rich” sugar molecules). A well-aligned activity could have students use diabetic strips to show that sugar is present in Iris leaves that have been grown in the light, but not in the dark. Another well-aligned activity compares a plant’s leaf to a factory in terms of inputs, outputs, and energy source. On the other hand, activities that show that a plant deprived of light does not grow well or that plants grow toward a source of light demonstrate that plants respond to light and may need light to grow (ideas found in elementary school benchmarks), but not that light energy is needed for the production of sugars in plants and that those sugars store some of the energy. Similarly, separating leaf pigments on paper chromatography or reading an explanation of how chlorophyll molecules can be excited to a higher energy configuration by sunlight and in turn excite molecules of carbon dioxide and water so they can link are more sophisticated than the idea that “plants use light energy to make energy-rich sugars” (Stern & Roseman, 2001). The analysis of content alignment does not conclude with a score; recording all relevant evidence from the materials, reviewers report on whether the material aligns with each key idea. These chunks from the materials are then analyzed using the instructional criteria.

Instructional Support. Including content that is aligned with worthwhile key ideas is an important first step, but is nonetheless inadequate for learning. For learning to take place, effective instructional strategies must be consistently aimed at the key ideas (NRC, 2000; Posner et al., 1982). The Project 2061 curriculum analysis procedure examines the quality of instructional support materials provide for learning and teaching the key ideas. Criteria for judging the quality of instructional support focus on many aspects of instruction and are derived from the learning sciences research. This includes criteria that judge whether the material conveys a sense of purpose to students or suggests how teachers might do so (Hart, Mulhall, Berry, Loughran, & Gunstone, 2000; Wise & Okey, 1983), whether it helps teachers to take account of students’ prior ideas (e.g., Lee et al., 1993), whether it includes vivid experiences with relevant phenomena that might make the scientific ideas plausible (Posner et al., 1982), and whether it provides opportunities for students to use their ideas (NRC, 2000). Materials are judged by what is explicitly included in them—not by what exemplary teachers might be able to do to improve them. For each criterion, reviewers cite relevant evidence from the student text or the teacher’s guide, and using predetermined scales, rate each criterion as 3.0 (*excellent*), 2.5 (*very good*), 2.0 (*satisfactory*), 1.5 (*fair*), or 0 to 1.0 (*poor*). This rating process is illustrated in the next section for the criterion, “Providing variety of phenomena.” The procedure is described in detail in Roseman et al. (1996), Kesidou and Roseman (2002, 2003), and at www.project2061.aas.org (Research studies that focus on the specific criteria that are discussed in this article are cited later; research base for additional criteria can be found in Kesidou & Roseman, 2002.) Appendix A lists all instructional criteria, and Appendix B includes the clarification and scoring guideline for the criterion “Providing variety of phenomena.”

Assessment Quality. With respect to assessment in curriculum materials, judgments are made on the basis of (a) whether assessment questions and tasks appear to aim at specific benchmarks and standards and are likely to reveal what students’ actually know (as opposed to rote memorization of these goals) and (b) whether assessment embedded in curriculum materials throughout instruction can be used for making modifications in instruction. The specific criteria to

evaluate assessment together with the indicators of meeting them and a relevant literature review are presented in Stern and Ahlgren (2002). In addition, findings related to assessment quality, a range of examples of assessment tasks that meet (or do not meet) each of the indicators and criteria, and examples of how reviewers arrived at particular scores will not be described here but are presented elsewhere (Stern & Ahlgren, 2002).

The Review Process. As in other disciplines, in life science two independent 2-member teams trained in the use of the Project 2061 evaluation procedure independently analyzed the content alignment, quality of instructional support, and assessment in each curriculum material (Kesidou & Roseman, 2003). Analysts had the appropriate expertise in biology, and included experienced middle-school classroom teachers and science education university faculty members who were knowledgeable about cognitive research. The training focused on (a) clarification of the key ideas that served as the basis of analysis, (b) clarification of the meaning of the analysis criteria and indicators, and (c) applying the analysis criteria to analyze materials to ensure that the reviewers interpret key ideas and criteria correctly and support each of their judgments with evidence from the material (Kesidou & Roseman, 2003). After each team completed its analysis, reports were reconciled whenever possible to obtain a single judgment (Kesidou & Roseman, 2002). A complete record of ratings justified by evidence from the materials is available for each textbook (www.project2061.aas.org). The consistency of interanalyst scores was studied before this evaluation. In brief, when seven highly trained 2-member teams independently evaluated a middle-school science material for its treatment of the life science ideas, 87% of their scores were in agreement (Kesidou & Roseman, 2002).

This article describes the findings in two main areas: (a) content alignment to the life science ideas and (b) the instructional support the life science ideas receive for four of the criteria that served as the basis of the instructional analysis (i.e., alerting teachers and addressing commonly held students' ideas, providing a variety of phenomena, and including relevant representations).

Results

This section is divided into two parts. The first part reports on how well the content included in curriculum materials is aligned with the key life science ideas that served as the basis for our curriculum analysis. The second part reports on how well the materials support the learning of these ideas. Typical strengths and weaknesses in materials are described (using examples) to illustrate how reviewers arrived at particular scores.

Part A: Analysis of Content Alignment: Does Textbooks' Content Align with Key Ideas Related to Matter and Energy Transformations in Living Systems?

Eight of the nine textbook series include sections that relate to flow of matter and energy, such as food making in plants, digestion in animals, cellular processes for releasing energy, energy flow in food webs, and more. Furthermore, the middle-school materials examined in this study treat most of the *specific* key ideas that served as the basis for the analysis. Two materials (*Science 2000*, and to some extent, *Science Interactions*) take on these ideas at the molecular level—i.e., matter transformation is presented in terms of the combination and recombination of atoms whereas other materials treat the key ideas in terms of the transformation of *substances* (Either approach might be appropriate for middle school, according to *Benchmarks for Science Literacy* or the NSES). However, while the ideas are included in materials, it appears that materials often focus on the

processes of photosynthesis and respiration in terms of inputs (reactants) and outputs (products), and not on the idea that substances are being *transformed* to other substances. In addition, a few key ideas—such as the idea that food is a source of both building blocks and energy—are rarely present in materials. These findings are elaborated next.

One material, *Middle-School Science & Technology*, treats so few of the key ideas that it was not analyzed further for instructional quality. However, this material does include two vivid experiences related to the key idea that decomposers break down dead organisms into simpler, reusable substances and includes some text statements related to a few other key ideas.

The Dual Role of Food (Rather than Food Just Needed for Energy)

A few key ideas (or parts of ideas) are rarely treated in middle-school materials. For example, typically there is not a match in these materials to the idea that food serves not only as fuel but also as a source of building blocks for all organisms. Materials may discuss food only as an energy source, ignoring its important role as building material (e.g., *SciencePlus*), focus on the dual role of food mostly for humans, hardly mentioning other organisms (e.g., *Prime Science*), or not attempt to teach this idea at all (e.g., *Glencoe: Life Science*). Another concern is with materials (e.g., *Science Interactions*) that postpone the treatment of what food is until late in the program—after the processes of food making and breaking down are already discussed. Only a few materials appear to acknowledge the fundamental importance of this idea by taking it on before treating ideas that depend on it (e.g., *Science 2000*, Grade 6, Unit 4, Cluster 25).

Transformations of Matter and Energy (Rather than Just Reactants and Products)

Benchmarks for Science Literacy (AAAS, 1993) and the NSES (NRC, 1996) recommend that middle-school students should understand metabolic processes not just in terms of initial and final substances but that in these processes the initial substances are *transformed* into the final substances, with the total amount of matter being constant in the process. Likewise, the literacy expectation with respect to energy is that, as in physical systems, energy in living systems may change forms but is not created or destroyed. In contrast, most materials do not focus on the transformations of matter and energy. Materials rarely treat transformations of *matter* (e.g., the assembly of sugars made during photosynthesis into the plants' body structures or the breakdown of the sugars into carbon dioxide and water). Instead, the focus, both during instruction and in assessments, is on naming the reactants and products of photosynthesis and cellular respiration or on comparing the two processes.

Similarly, instead of considering the energy transformations during photosynthesis and cellular respiration, most materials state, explain, and provide activities for the idea that light energy is *necessary* for the making of sugars in plants. However, the idea that some of the light energy is transformed into “chemical” energy in the newly made sugars (as evidenced by the fact that we can get some of it back) is usually ignored (e.g., *Prentice Hall*, *SciencePlus*, *Glencoe: Life Science*). Consequently, students might infer that light energy is a necessary “ingredient” for photosynthesis to take place and that it is “used up” like carbon dioxide and oxygen (The fact that many textbooks present light as a “reactant” in the photosynthesis equation, without indicating that the products are “energy rich,” fosters this notion.) Alternatively, students might think that light is a “facilitating agent” (just as light makes it possible for people to, say, read a book, so light makes it possible for plants to make food) rather than that light energy is converted to another form that is stored. When materials introduce the concept of releasing energy from sugars (in plants or other organisms), they typically do not point out that the energy in sugars comes from light energy

originally captured by the plant. Since many students have difficulties appreciating that energy cannot be created or destroyed, they may think that the light energy “disappeared” and that glucose energy is a “new,” unrelated energy.

Connected Versus Unconnected Presentation of the Key Ideas

While *Benchmarks* (AAAS, 1993) and the NSES (NRC, 1996) specify what students ought to know by the end of particular grade ranges, they do not dictate *when* during the grade range a given benchmark (or standard) should be taught or in what sequence the set of ideas should be presented. In life science materials, both those in which life science topics are taught in a single grade and those in which the topics are distributed over Grades 6 to 8, key ideas about matter and energy transformations are treated in different chapters that deal with cells, organisms, and ecosystems. For students to see individual presentations as instances of matter and energy transformations and to see connections among the key ideas, materials need to make these connections explicit. Unfortunately, most textbooks fail to make the needed connections. Ideas about matter and energy transformations at the different levels of biological organization are rarely connected to one another. For example, in two materials (*Prime Science* and *Prentice Hall*), decomposition and respiration are presented in different chapters. When decomposers are introduced, these materials just state that decomposers break down dead organisms into simpler substances, but do not specify what these “simpler substances” are or that decomposers simply carry out respiration to obtain energy and build their body structures. Similarly, few textbooks connect the idea that plants make their own food to the idea that some of the sugars made by plants are assembled into the plant body structures. Textbooks (e.g., *Prentice Hall* and *Glencoe: Life Science*) have a section on plants’ parts—separate from the one on photosynthesis—in which examples are given of plants that store food in their roots and stems; however, the textbooks do not note that this stored food comes from the food that the plant made during photosynthesis.

Part B: Analysis of Instructional Quality: Do Textbooks Support Learning and Teaching of the Key Ideas Related to Matter and Energy Transformations in Living Systems?

This part first presents the scores curriculum materials received on *all* the instructional criteria and then elaborates on findings related to three main instructional areas:

1. the extent to which curriculum materials take account of students’ prior knowledge (Category II, Appendix A),
2. the extent to which curriculum materials introduce sufficient relevant phenomena (Category III, Appendix A) and,
3. the extent to which curriculum materials include helpful representations (second criterion in Category IV, Appendix A).

While in general, eight of the nine materials include content that matches the key ideas, the instructional support for these ideas is minimal. Table 1 shows how well *each* of the textbook series evaluated in this study scored on each of the instructional criteria. Table 2 presents the *average* scores and range for the quality of instructional support for the textbooks evaluated. As shown in the tables, currently available textbooks do not support the attainment of key ideas about flow of matter and energy in living systems recommended by national standards documents. Science curriculum materials include inadequate support for learning and inadequate ways to probe students’ understanding (as suggested by the scores of Category VI: Assessing Progress). With the

Table 2
*Average scores of instructional quality in student and teacher editions
 (textbooks in alphabetical order)*

Textbook Series	Average	Range
<i>Glencoe: Life Science</i>	.62	0–1
<i>Macmillan McGraw-Hill Science</i>	.79	0–1.5
<i>Middle-School Science & Technology</i>	No match to life science ideas	
<i>Prentice Hall: Exploring Life Science</i>	.34	0–1.5
<i>PRIME Science</i>	.61	0–1.5
<i>Science 2000</i>	1.09	0–2.5
<i>Science Insights</i>	.33	0–1
<i>Science Interactions</i>	.63	0–1.5
<i>SciencePlus</i>	1.09	0–2.5

0.1 = poor; 1.5 = fair; 2 = satisfactory; 2.5 = very good; 3 = excellent.

exception of two materials, all other materials received below satisfactory scores on all 19 criteria. Even *SciencePlus* and *Science 2000*, the highest rated materials, received a satisfactory score on only four and three of the criteria, respectively, and their average score for the quality of their instructional support for learning the key ideas is only 1.09 of 3.0. When comparing the scores of materials in the different categories on which the instructional analysis is based—Identifying and maintaining a sense of purpose, Taking account of student ideas, Engaging students with phenomena, Developing and using scientific ideas, Promoting student thinking, and Assessing progress—they are evenly low in all categories. In the following, the main flaws in materials—namely, not taking into account student ideas, lack of appropriate phenomena, and lack of helpful representations—are described and exemplified.

Since textbook series do not provide sufficient relevant phenomena that could make the ideas chosen for this study plausible nor meaningful representations that could make these ideas intelligible, discussion of additional criteria is less important. In the absence of appropriate phenomena or comprehensible representations, there is little basis, for example, for students to express or explain their ideas and no basis for guiding student interpretation and reasoning. Likewise, reporting on the findings about the practice tasks included in these materials is not crucial because—even if very thoughtful tasks are included—students have little basis to respond successfully.

Curriculum Materials Do Not Take Account of Student Ideas (Category II, Appendix A)

Fostering understanding requires taking the time to attend to the ideas students already have, for example, to inform teachers about prerequisite ideas and about students' probable conceptions, and to incorporate appropriate strategies to address students' ideas (Bishop & Anderson, 1990; Eaton, Anderson, & Smith, 1984; Lee et al., 1993; McDermott, 1984). Currently available curriculum materials do not attend to important prerequisites before introducing more sophisticated ideas. For example, some materials use chemical formulas, taking for granted that students can make sense of them. Many materials do not clarify the scientific meaning of the word "food"—that is, substances from which organisms derive the energy they need to carry out life processes and material of which they are made—that is very different from the everyday meaning of this word and refers to whatever nutrients organisms *take in*. Students who view food as anything "taken in" may not appreciate the distinction made in the key idea that "plants make

their own food whereas other organisms consume food” since plants do take in water and minerals from the soil. Despite the fact that this important stumbling block has been discussed and exemplified in the literature (e.g., Anderson et al., 1990), most materials do not take advantage of this information. Some materials present prerequisite ideas, but not in time for them to be helpful. *Science Interactions*, for example, states that the energy for the photosynthesis reaction comes from sunlight in Grade 6 (Course 1, p. 336) materials, but it is not until Grade 7 that students are told that energy exists in different forms (Course 2, pp. 138–140). Without an appreciation that energy can exist in different forms and can be changed from one form to another, students may think that light is “used up” in photosynthesis.

As described in the Introduction of this article, research studies conducted over the past 2 decades indicate that students have difficulties understanding ideas about matter and energy transformations, matter conservation in living systems, and plant and animal nutrition. Furthermore, these particular misconceptions are especially resistant to change (Anderson et al., 1990; Bell & Brook, 1984; Roth & Anderson, 1987). For meaningful learning to occur, teachers need to be informed about commonly held ideas students might have and about what is likely to work. However, currently available materials provide little support. While many curriculum materials include sections with titles such as “Prior Knowledge and Misconceptions” or “Addressing Misconceptions” in the teachers’ guides, they rarely alert teachers to the relevant, commonly held student ideas extensively documented in the research literature. A few materials include general statements that encourage the teacher to find out whatever misinformation students bring to the classroom, but provide no information on what to look for. Other materials highlight possible student preconceptions, but the preconceptions highlighted are not those identified by research nor are the preconceptions highlighted particularly relevant to the transformations of matter and energy in living things. For example, one material (*Science Interactions*) points out that “students may think that all plants have roots, stems, leaves, and vascular tissue” (Course 1, p. 317) and that “Some students may find it difficult to master the definitions of many terms presented” (Course 1, p. 324), but does not inform teachers that many students think that plants get their food from the soil (Roth & Anderson, 1987; Wandersee, 1985). In the few instances where textbooks do alert teachers to specific, relevant commonly held ideas (*Prime Science* and *Macmillan*), they do not adequately explain them. For example, *Prime Science* mentions that energy flow and loss are difficult ideas for students to comprehend (Level 1, Vol. II, chap. 12: Balancing Acts, p. 828), but does not explain what makes these concepts difficult for students and how it impacts learning ideas about energy transformation.

Most importantly, curriculum materials provide almost no support for addressing these naïve conceptions. Materials usually do not include questions designed to engage students in contrasting their ideas with other ideas or to challenge specific, common student misconceptions. Even when potentially helpful tasks are included, teachers are not alerted to the role they could play in addressing common misconceptions nor do the materials themselves attempt to do so. For example, *SciencePlus* includes an activity that could help address the misconception that plants get their food from the soil (Level Blue, pp. 11–12), but does not use it to do so: Students read about van Helmont’s famous experiment that showed that a plant’s mass does not come from the soil. However, after reading that (a) the soil lost only a few grams even though the willow tree grown in it gained over 70 kilograms and (b) that Van Helmont concluded that the willow tree’s increase in mass came from the water (since that was the only ingredient Van Helmont added), students are just asked what was the mass of the earth and the tree in the experiment, whether van Helmont was a good experimenter, and what was wrong with his conclusion (p. 12). The teacher’s guide gives the correct answers, but does not encourage students to link the experiment to their own ideas about where plants get their food nor does it inform teachers about students’ commonly

held ideas regarding plant nutrition to facilitate such a link. Teachers who are not already aware that students think plants take in food from the soil may gloss over this activity.

In contrast, *Food for Plants* (pp. 37–40), a stand-alone unit developed through Research and Development efforts at Michigan State University (Roth, 1997), uses the same experiment more effectively by having students first write down and justify their ideas about whether they think soil is food for plants (Teacher's Guide, p. 39), next predict what would happen to the weight of 200 pounds of food and to the weight of a child eating that food (p. 38), then make predictions about the outcome of van Helmont's experiment (Teacher's Guide, pp. 40–41). After they read about van Helmont's findings, students are again encouraged to think about whether soil is food for plants. Teachers are informed that the purpose of the activity is to challenge students' ideas about plants' food (Teacher's Guide, p. 39) and are encouraged to emphasize "the contrast between the students' predictions and what actually happened in the experiment" (Teacher's Guide, p. 42). These examples are included in Appendix C.

Curriculum Materials Do Not Provide Sufficient Phenomena to Make Key Ideas about Matter and Energy Transformations in Living Things Plausible to Students (Category III, Appendices A & B)

Much of the point of science is explaining phenomena in terms of a small number of principles or ideas. For students to appreciate this explanatory power, they need to have a sense of the range of phenomena that science can explain. Appropriate phenomena, introduced directly through hands-on activities or demonstrations or indirectly through the use of videos or text, can help students view scientific ideas as plausible (Anderson & Smith, 1987; Champagne, Gunstone, & Klopfer, 1985; Strike & Posner, 1985). Seeing that a variety of phenomena can be explained by an idea is critical to appreciating its explanatory power and finding it plausible. For example, to help students see the idea that plants make their own food from carbon dioxide and water as plausible, materials could involve students (in activities or readings) in observing that sugar is present on leaves that have been grown in the presence of CO₂—but not in its absence—or that sugar (or starch) can be detected on leaves grown in open jars, but only in the first few hours on similar leaves grown in closed jars. Moreover, for students to make important generalizations, they need to encounter ideas in a variety of instances. For example, students who observe that a potted plant makes sugars only in the presence of carbon dioxide may not appreciate that the same holds true for trees, bushes, grass, flowering plants, or algae.

This study revealed that middle-school materials do not include a sufficient number and variety of phenomena relevant to the set of key ideas about matter and energy transformations in living things, as reflected in uniformly poor ratings on this criterion (see Table 1). The scoring scheme for this criterion was based on the average scores of the number and variety of phenomena provided for each key life science idea (see Appendix B). As a result, a poor score may represent different findings in different materials: Some materials include almost no phenomena that are relevant to the key life science ideas; other materials do provide a few sound phenomena, but do not do so systematically for all key ideas. In both cases, the number and variety of phenomena that support the set of key life science ideas were judged to be inadequate. For example, *Science 2000* provides the following phenomena for the idea that food serves as fuel and building blocks: Students examine food labels in terms of calories and main ingredients (Grade 6, Unit 4, Cluster 25, T10 and T12), they test different foods for sugar, starch, fat, and protein (Grade 6, Unit 4, Cluster 25, 25-1-B), they observe that more heat is given off when they burn buttered popcorn than unbuttered popcorn (Grade 6, Unit 4, Cluster 25, T12), and they watch a video showing a toy engine and a light bulb powered by burnt peanuts (Grade 6, Unit 4, Cluster 25-2, Video: *Measuring*

the Energy Stored in Food). Even though there are no phenomena that show that *all organisms* (as opposed to only humans) get their energy and building matter from their food, *Science 2000* provides the most extensive set of phenomena for this idea compared to other materials. On the other hand, *Science 2000* does not provide nearly enough phenomena to support the other key ideas, resulting in a “poor” overall rating for this criterion.

While all materials attempt to include hands-on activities, these often are not focused on the key ideas examined here. In analyzing phenomena directly observed or read about, we have found the following:

- *Phenomena that do not align with key ideas.* For example, relevant to the idea that “plants use energy from light to make energy-rich sugars,” materials often have students observe that starch is present in leaves grown in the light, but not in the dark. While this phenomenon provides direct support for this key idea, other phenomena are included that are only peripheral to the key idea. For example, observations that plants cannot grow well without light (target the less sophisticated idea that plants *need* light), that the rate of photosynthesis increases as the light intensity increases (target a more sophisticated idea that goes beyond the middle-school literacy expectation), or that green leaves are composed of several pigments (target the general topic, photosynthesis, but not any of the key ideas on matter and energy transformations).
- *Phenomena that align with key ideas, but are not adequately linked to them.* Students’ attention is not always drawn to the connection between their observations and the relevant scientific ideas. For example, in one material (*Macmillan*, *The Plant Kingdom*, pp. 34–35), an activity that has students observe that in the presence of light, Elodea plants “remove” carbon dioxide from the solution in which they are grown, could be used to support the idea that plants use the energy from light to make sugars from carbon dioxide and water; however, neither matter nor energy transformation is emphasized since neither sugars nor light energy are mentioned. Hence, it is unlikely that the activity would make the idea plausible, and so the material was not given credit for this activity. Another material, *Science Interactions* (Course 2, p. 596), has students record temperature changes of soaked beans and dry beans; however, this activity is linked only to the term “respiration,” assuming students understand what it means, but not to the idea that the beans obtain energy stored in the sugars, releasing some of it to the environment as heat.
- *Linking some phenomena to key ideas requires students to draw conclusions based on insufficient evidence.* Unfortunately, this was the case for many materials. For example, relevant to the idea that organisms break down the sugars into simpler substances, *Prentice Hall: Exploring Life Science* (Teaching Resources, chap. 3) has students incubate yeast and sugar, smell the alcohol produced, and observe color change, which indicates carbon dioxide production; however, since this experiment is not controlled and students do not observe what happens when sugar (or yeast) are *not* added, it is unlikely that the observations will make the idea any more plausible for them. Or, in another material (*Science Interactions*, Course 1, p. 336), students blow into bromothymol blue solution and observe the change of color. Then, they add Elodea plants to the beaker, incubate it under bright light, and record color change. Students are expected to infer that plants remove carbon dioxide from the solution; however, the experiment is not controlled properly, and as far as students are concerned, the bright light could have caused the color change. In yet other materials (e.g., *SciencePlus*, Level Blue, p. 14), students test leaves for starch and detect starch only in leaves that were grown in the presence of carbon dioxide. They conclude that carbon dioxide is necessary for green plants to make starch. While this activity is helpful, students are not told that iodine turns black *only* in the presence of starch and not in the presence of any other substance, and hence the conclusion might seem invalid for them.

- *Some phenomena are used to teach experimental design skills rather than a key idea, so the link to the scientific concept may not be made.* For example, relevant to the idea that plants make sugars from carbon dioxide and water and that in the process light energy is transformed and stored in sugars, *Prime Science* (Level C, Green Machine, p. 5 and Teacher's Guide, pp. 51–52) has students plan their own experiments to test the effects of light, carbon dioxide, and temperature on the rate of photosynthesis. However, it is not clear whether students will even carry out the experiments they design since the point seems to be to teach experimental design. If students do not carry out the experiments, they will not observe the phenomena (and clearly, the link to the scientific idea will not be made).
- *Some experimental setups required to observe the phenomena seem overly complex.* For example, relevant to the idea that plants make sugars from carbon dioxide and water, *Science 2000* (Unit 2, Cluster 12, Lessons 12-1–12-5) involves students in an investigation in which students observe what happens to plants grown in a variety of conditions: in the absence of water, in the presence of sodium hydroxide (which absorbs carbon dioxide), in the light, in the dark, in the refrigerator, and so forth. Each group of students performs only parts of this investigation and reports findings to the whole class. In some cases, one group tests a certain variable (e.g., dark) while a different group performs the control experiment (e.g., light). The investigation continues over several lessons, but it is unclear whether and how students will be able to assemble the various findings and consider them in light of key ideas.

Curriculum Materials Do Not Provide Adequate Representations to Make the Key Ideas Intelligible for Students (second criterion in Category IV, Appendix A)

Multiple representations such as diagrams, models, simulations, or analogies can help make abstract scientific ideas intelligible to students (Champagne et al., 1985; Feltovich, Spiro, Coulson, & Anderson, 1989; Stavy, 1991; Strike & Posner, 1985). Different representations highlight different aspects of an idea and provide a variety of opportunities for the idea to connect to other students' ideas and become embedded in a student's knowledge system. Nonetheless, none of the materials examined adequately represent the key ideas. Furthermore, representations that are included are often incomprehensible. It is common to see in materials illustrations of processes that are quite complex, but insufficiently explained (e.g., a photosynthesis diagram in *Science Insights*, p. 98), diagrams filled with technical terms that have not been previously introduced (e.g., a leaf cross section in *Prentice Hall: Exploring Life Science*, p. 233), chemical equations for photosynthesis and respiration without prior attention to chemistry prerequisites (e.g., *Glencoe: Life Science*, pp. 318, 321), or food web diagrams in which switching the color code in the middle of the diagram or from one diagram to the next undermines students' ability to track the flow of matter or energy (e.g., *Science Interactions*, Course 1, pp. 336, 341; *Glencoe: Life Science*, p. 498).

Some materials attempt to draw analogies that are potentially helpful, but they fail to provide adequate help for students or guidance for teachers to use with students to make the analogies useful. For example, the teacher's notes in *Macmillan* (Earth's Ecosystems, p. 20) suggest that students will compare the recycling of resources by humans with that in nature or suggest to discuss with students examples of cycles (seasonal change, menstrual cycle, etc.); however, no other guidance is provided for the teacher to develop these analogies and discuss with students how these cycles are similar and yet different from cycles in ecosystems. Other materials attempt to draw analogies between plants and food factories or power plants (*Science Interactions*, Course 2, p. 594; *Prentice Hall: Exploring Life Science*, pp. 82, 233, 236; *Science Insights*, p. 243) or

between the release of energy from food and the burning of gasoline (*Science Interactions*, Course 2, p. 564; *SciencePlus*, Level Red, p. 28); however, these analogies, like those just noted, are inadequately explained. Students' attention is not drawn to the similarities and differences between a plant cell and a factory or between releasing energy from food and burning. Surely, analogies can be helpful to clarify abstract ideas only if students are already familiar with the analogy (Thagard, 1992) or if it is explained to them. For students not previously familiar with burning, power plants, or recycling, these analogies might be almost as foreign as the concepts they attempt to represent. It seems, however, that curriculum materials take for granted that students are familiar with the analogies employed.

While representations typically highlight only some aspects of an idea and therefore might represent reality well in some aspects and not very well in others, middle- and high-school students tend to see models as an actual copy of reality and not as conceptual representations (Grosslight, Unger, & Jay, 1991). Because of this potential pitfall, care must be taken that representations included in textbooks will represent the real thing as accurately as possible—or that they involve students in considering which aspects of the real thing are represented by the model and which are not (Thagard, 1992).

Unfortunately, many textbook series include representations that could actually mislead students. For example, relevant to matter transformations, three materials include a diagram of the carbon dioxide–oxygen cycle through photosynthesis and respiration that could mislead students to think that carbon disappears in photosynthesis and reappears in respiration (as opposed to being combined and recombined in organic molecules) (*Science Interactions*, Course 1, pp. 336, 341; *Macmillan, Earth's Ecosystems*, pp. 20–21; *Prime Science*, Level 1, p. 334). These materials name or show the chemical formulas of oxygen and carbon dioxide, but ignore sugars. Relevant to energy transformations, some materials include the word equations of photosynthesis and respiration, showing light energy as one of the reactants or the products (e.g., *Science Insights*, p. 98; *Glencoe: Life Science*, pp. 318, 321). Students might conclude from such diagrams that energy can change into matter or simply disappear. *Science 2000* (Grade 6, Unit 4, Cluster 25-2, p. T-15, "Elements and our body") includes a diagram showing the relative *proportions* of different elements in the body, but the proportions are superimposed on a diagram of a human body and might mislead students to think that they represent the *distribution* of these elements in the body.

One of the most misleading representations found is a diagram of a chloroplast and a mitochondrion connected by arrows in a cycle indicating that O_2 and glucose are produced in the chloroplast during photosynthesis and transferred to the mitochondrion, which in turn produces CO_2 and H_2O during respiration, which are transferred to the chloroplast (*Glencoe: Life Science*, p. 79). The diagram is accompanied by the balanced chemical equations of photosynthesis and respiration, and the text points out that the products of one process are exactly the reactants of the other. Students might be misled to think that these substances are cycled in a plant cell and balance each other off so that there is no net production of any substance. Nowhere does the text clarify that the rate of photosynthesis is far more than the rate of respiration, and that this is the reason why the net effect of photosynthesis is to produce enough food (and oxygen) for both plants and the organisms that consume them.

Of course, since representations can highlight only some aspects of an idea or process, no representation can be completely accurate. Materials could address this issue by asking students to consider how the representations included are like and unlike the thing being represented (Thagard, 1992). None of the materials used this strategy. In addition, while no *single* representation can be completely correct, a *variety* of representations of the same idea, each representing different aspects of the idea, might contribute to a more complete understanding (Feltovich et al., 1989; Spiro, Coulson, Feltovich, & Anderson, 1994). Indeed, this is why multiple

representations are used in science. As reported, however, such variety was not present in any of the materials.

Only two materials (*Sciences Interactions*, Course 2, p. 601, and more elaborated in *Science 2000*, Grade 6, Unit 4, Cluster 25-2, p. T-16) include a representation that might help to make the abstract idea of matter transformations more intelligible. Students are asked to count the atoms on both sides of the respiration equation and to decide whether there were any atoms created or destroyed during respiration. But even these potentially helpful representations are not followed by sufficient questions or discussions to foster the notion that atoms of molecules are rearranged to make new substances.

Summary and Discussion

The study described in this article finds that currently available materials are not likely to contribute to the attainment of benchmarks and standards related to matter and energy transformations in living systems. In terms of content alignment, eight curriculum materials (of the nine examined) devote a significant number of pages relevant to the topic flow of matter and energy, but the textbooks do not emphasize the specific key life science ideas that served as the basis for analysis. Most key ideas are introduced several times in each of the materials, but are often buried between unrelated ideas (e.g., classifications of living organisms or plant parts) or even more sophisticated ideas, making it difficult for students to focus on the main ideas. Moreover, the ideas about matter and energy transformations are merely repeated in several contexts rather than being explicitly extended to new contexts or revisited in progressively higher levels of sophistication. Most importantly, the instructional support tied to the key life science ideas is minimal.

It may be argued that textbooks did not do well in this study simply because the key ideas that served as the basis of our analysis were not at the focus of the instruction. However, the amount of textbook space devoted to this topic is enormous: No other topic area in life science received more attention than the flow of matter and energy (Kesidou & Roseman, 2002, 2003). Moreover, these curriculum materials received overall low ratings on their treatment of the other two science topics examined in Project 2061's study: namely, the kinetic molecular theory and processes that shape the Earth (Caldwell & Stern, 2000; Kesidou & Roseman, 2002). Since these three topics have been included in textbooks for many years, it is unlikely that materials would have received higher scores had key ideas on other topics from *Benchmarks for Science Literacy* (AAAS, 1993) and the NSES (NRC, 1996) been selected.

Benchmarks for Science Literacy (AAAS, 1993) portrays a rational progression of ideas across the K–12 grades that lead to science literacy in specific topics. Understanding of matter and energy transformations in ecosystems starts in primary school with awareness of the basic components of the food chain and with the tracing of chains of what eats what in various environments. In middle school, these ideas lead to the more advanced understanding of food chains in terms of transformations of energy and transformations of substances into other substances in the food web. Finally, in high school, these ideas are recast in terms of rearranging atoms of the original substances into new molecules and are connected to energy transformation with the admittedly sophisticated idea that changing configurations of atoms in molecules absorbs or releases energy (AAAS, 2001a). Transformations of matter and energy are therefore a needed intermediate step between the simple notion of food chains (i.e., what eats what) and the more complex understanding of photosynthesis and respiration across the different biological organization levels. Indeed, some research identifies two major ways of thinking about photosynthesis and ecosystems: one in which ideas of consumption and production dominate and

one more sophisticated way which incorporates understanding of matter transformation (Carlsson, 2002a,b).

Unfortunately, as shown in the previous section, materials' treatment of photosynthesis and respiration often focuses on naming reactants and products rather than on the concept that matter (and energy) are being *transformed* into other substances (or other forms of energy) (e.g., the conversion of carbon dioxide and water into sugars or the assembly of sugars made during photosynthesis into plant structures such as roots, leaves, flowers, seeds, etc.). In the absence of an emphasis on transformation, students may see plants (or any other living organisms) as taking in and using some substances and producing some others as separate events, not appreciating that the substances taken in are raw materials for the products.

Transformations of matter and energy may seem too ambitious for concrete operational learners and hence might not be developmentally appropriate for middle-school students (Shiland, 2003; Simpson & Marek, 1988). However, some research indicates that carefully designed instruction can help even sixth-grade students develop an understanding of atoms and molecules (Lee et al., 1993) and photosynthesis (Roth, 2001). These studies suggest that if effective instruction were applied towards the end of the middle-school grade range, it would probably result in even better learning, particularly if coupled with the use of good models and sound phenomena. In addition, it had been argued that the type of educational experiences students have can influence the amount of time required for them to pass through one stage into another (Renner & Marek, 1990). Therefore, we suggest that the persistence of some commonly held ideas reported in the literature is not necessarily the result of a lack of reasoning abilities of learners (or age inappropriateness) but more likely the result of poor curricula and instruction (Stern, 2003). For example, as shown in the previous section, textbooks fail to tie together different relevant experiences that students have with the key ideas: The food-making process, for instance, is often treated in one chapter whereas a different chapter (that might be separated by several unrelated chapters) focuses on plant anatomy and explains that food is stored in certain plants' parts. No explanation, however, is provided for where this stored food comes from and how it relates to the photosynthesis process. In neither place is the incorporation of sugars into the plant's body structures explained. When seventh- and eighth-graders—who had been taught the topic of photosynthesis a year before—were asked what would happen to the size of growing carrots if some animal would eat only the leaves, most said that the carrots will continue to grow to their normal size (Project 2061, unpublished data).² The fact that these students as well as many high-school and even MIT graduates fail to explain where the “stuff” in plants and trees comes from should therefore come as no surprise. These students clearly did not receive appropriate instruction to help them understand the relevant ideas or challenge their naïve conceptions. As indicated in our study, none of the currently available materials (including the highest rated ones) provide relevant phenomena to make the key ideas plausible or provide adequate representations to make these abstract ideas more intelligible for students. While most material developers emphasize hands-on or inquiry-derived activities in curriculum materials, including such activities does not guarantee learning of important ideas. We have identified activities that are not particularly relevant to the key life science ideas or not explicitly linked to them, involve overly complex experimental setups, require students to draw conclusions based on insufficient evidence, or focus on experimental skills rather than on key ideas. These findings are consistent with previous studies that noted that school laboratories often lack purpose and overburden students with many details to recall such as experimental setups and instructions (Hart et al., 2000; Johnstone & Wham, 1982).

Finally, despite the extensive, well-documented research on students' difficulties regarding food, matter and energy transformations, matter conservation in living systems, and plant and

animal nutrition, middle-school textbooks do not take these difficulties into account. And as argued in the previous section, text statements and representations in materials can often reinforce or even induce students' naïve conceptions. What we already know about student learning does not seem to influence materials' design.

Our overall findings are in accord with previous studies that looked at a variety of aspects of curriculum materials (Abraham et al., 1992; Schmidt et al., 1997; Tyson-Bernstein, 1988).

Implications and Recommendations

For better or for worse, the majority of schools are still relying on textbooks as the primary source of the classroom curriculum, and textbooks strongly influence student learning through their influence on teachers. While science educators have different views on whether curriculum materials are needed at all and on the role they should play, the underlying assumption of this study is that curriculum materials can and should play an important role in improving teaching and learning. Rather than trying to "teacher-proof" materials, we assume that materials that rate highly on our criteria support teachers by helping them to build their own content and pedagogical knowledge and to reflect this knowledge in their teaching. For this reason, the goal of this study was not only to evaluate what can be found in curriculum materials to support the learning of important ideas but also to characterize how to enhance this support.

To improve learning of all three topics chosen for this study, our findings indicate that critical and thoughtful steps need to be undertaken in developing new middle-school curriculum materials. *Science for All Americans* (Association for the Advancement of Science, 1989), *Benchmarks for Science Literacy* (AAAS, 1993), and the NSES (NRC, 1996) can provide materials' developers with a coherent set of learning goals on which to focus their materials, and the *Atlas for Science Literacy* (Association for the Advancement of Science, 2001a) can shed light on conceptual prerequisites and connections. The instructional criteria used in this study can provide further guidance for curriculum development by specifying and illustrating research-based qualities of instructional support. This support is defined by our criteria to mean establishing a sense of purpose for both students and teachers, taking account of students' prior ideas (both troublesome and helpful) and attempting to address these ideas, presenting sufficient relevant phenomena to make the scientific ideas more plausible, helping students to conceptualize ideas by including helpful representations, providing varied opportunities to practice the ideas, and guiding students' interpretation of what has been learned. Each criterion is defined by a unique set of indicators that further specify what constitutes evidence for meeting that criterion.

Concrete examples from curriculum materials, identified and described as a part of this study (Association for the Advancement of Science, 2001b; www.project2061.aaas.org), especially from a few research-based, stand-alone middle-school units developed at Michigan State University, can provide additional insights for curriculum development by illustrating characteristics of materials. For example, *Food for Plants* (Roth, 1997, 2001) takes on the idea that plants make their own food (sugars) from carbon dioxide and water. It attends to prerequisite ideas, alerts teachers to commonly held students' ideas, includes tasks that can help the teacher to find out what his or her *own* students know before instruction, attempts to address students' misconceptions regarding plants' food, and guides students' interpretation about their experiences. Questions used (and validated) in written questionnaires and interviews in high-quality research projects can serve as a source of assessment tasks to monitor student ideas before, throughout, and following instruction (e.g., Sadler, 1998; Wandersee, 1985; White & Gunstone, 1992). In fact, the criteria and considerations described in this article are now being used to guide curriculum development (McNeill et al., 2003) and to monitor the progress of this development in a long-term

partnership established between Project 2061 and some leading universities (for more details, see the Project 2061 Web site, www.project2061.org). It is hoped that the experience gained in this collaboration together with the criteria used in this study will be useful to commercial developers of instructional materials.

Will better curriculum materials necessarily make a difference in student learning? Clearly, teachers play a crucial role in mediating even the best-available curriculum and in modifying any curriculum as needed to meet their particular students' learning needs. In addition to specifications for designing new materials, the procedure and detailed analysis reports can be helpful to science teachers in providing concrete opportunities for teachers to reflect on their teaching practices. Highly rated materials can model attributes of exemplary teaching (Ball & Cohen, 1996). Over the past several years, professional development efforts by Project 2061 and others have introduced the curriculum evaluation procedure to thousands of in-service science teachers (Brearton & Shuttleworth, 1999; Kesidou, 2001). In addition, some teacher educators have begun to incorporate this tool into preservice education courses to make prospective teachers aware of the criteria when they design their lessons and supplement their textbooks (Hammrich, 1997; Lynch, 1999).

Finally, it is hoped that the Project 2061's evaluation procedure and the publication of these results will stimulate empirical tests of students' learning. For example, a number of phenomena—from the middle-school textbooks examined in this study and other sources—that could be used to make the ideas of matter and energy transformations in living systems plausible were identified in this study. Most of these phenomena have not been tested for their effectiveness in helping students learn these ideas. Will these and other phenomena indeed help students to view the key ideas as plausible? Will better representations help students understand these ideas better?

Additionally, in most materials matter is presented in terms of *substances* such as sugars, carbon dioxide, and water while the *molecular* explanation (i.e., combination and recombination of invisible particles) is not provided. However, research indicates that without a molecular explanation, the transformation of substances may remain mysterious for students, and they may see it as alchemy (Driver et al., 1994, p. 86). Clearly, more research is needed in the context of ecosystems to study the relative benefits of stopping with substance transformation as opposed to providing the full molecular explanation.

Notes

¹Findings related to materials' treatment of fundamental ideas in physical and Earth science are discussed elsewhere (Caldwell & Stern, 2000; Kesidou & Roseman, 2002).

²During 1999, a written questionnaire was administered to 300 7th- and 8th-grade students from various schools in Philadelphia. These students were taught the topic of photosynthesis earlier. When asked what would happen to the size of growing carrots if an animal would eat only the leaves, 70% of the students asserted that the carrots will continue to grow to their normal size. Of the students who thought that the carrots would be affected (30%), very few could explain why this would happen.

APPENDIX A: CRITERIA FOR EVALUATING THE QUALITY OF INSTRUCTIONAL SUPPORT

Project 2061's curriculum analysis procedure uses the following criteria, organized into seven categories, to determine the extent to which a material's instructional strategy is likely to help students learn the content. Each question is to be answered with regard to specific learning goals, not just in general.

Category I: Identifying and Maintaining a Sense of Purpose

Conveying unit purpose. Does the material convey an overall sense of purpose and direction that is understandable and motivating to students?

Conveying activity purpose. Does the material convey the purpose of each activity and its relationship to others?

Justifying activity sequence. Does the material include a logical or strategic sequence of activities (versus just a collection of activities)?

Category II: Taking Account of Student Ideas

Attending to prerequisite knowledge and skills. Does the material specify prerequisite knowledge/skills that are necessary to the learning of the benchmark(s)?

Alerting teacher to commonly held student ideas. Does the material alert teachers to commonly held student ideas (both troublesome and helpful) such as those described in *Benchmarks* chap. 15: “The Research Base?”

Assisting teacher in identifying student ideas. Does the material include suggestions for teachers to find out what their students think about familiar phenomena related to a benchmark before the scientific ideas are introduced?

Addressing commonly held ideas. Does the material explicitly address commonly held student ideas?

Category III: Engaging Students with Relevant Phenomena

Providing variety of phenomena. Does the material provide multiple and varied phenomena to support the benchmark idea?

Providing vivid experiences. Does the material include firsthand experiences with phenomena (when practical) and provide students with a vicarious sense of the phenomena when experiences are not firsthand?

Category IV: Developing and Using Scientific Ideas

Introducing terms meaningfully. Does the material introduce technical terms only in conjunction with experience with the idea or process and only as needed to facilitate thinking and promote effective communication?

Representing ideas effectively. Does the material include appropriate representations of the benchmark ideas?

Demonstrating use of knowledge. Does the material demonstrate/model or include suggestions for teachers on how to demonstrate/model skills or the use of knowledge?

Providing practice. Does the material provide tasks/questions for students to practice skills or *use* of knowledge in a variety of situations?

Category V: Promoting Student Thinking about Phenomena, Experiences, and Knowledge

Encouraging students to explain their ideas. Does the material routinely include suggestions for having each student express, clarify, justify, and represent his or her ideas? Are suggestions made for when and how students will get feedback from peers and the teacher?

Guiding student interpretation and reasoning. Does the material include tasks and/or question sequences to guide student interpretation and reasoning about experiences with phenomena and readings?

Encouraging students to think about what they have learned. Does the material suggest ways to have students check their own progress?

Category VI: Assessing Progress

Aligning to goals. Assuming a content match between the curriculum material and the benchmark, are assessment items included that match the same benchmark?

Testing for understanding. Does the material assess understanding of benchmark ideas and avoid allowing students a trivial way out, like repeating a memorized term or phrase from the text without understanding?

Informing instruction. Are some assessments embedded in the curriculum along the way, with advice to teachers as to how they might use the results to choose or modify activities?

Category VII. Enhancing the Science Learning Environment

Providing teacher content support. Would the material help teachers improve their understanding of science, mathematics, and technology necessary for teaching the material?

Encouraging curiosity and questioning. Does the material help teachers to create a classroom environment that welcomes student curiosity, rewards creativity, encourages a spirit of healthy questioning, and avoids dogmatism?

Supporting all students. Does the material help teachers to create a classroom community that encourages high expectations for all students, that enables all students to experience success, and that provides all students a feeling of belonging in the science classroom?

APPENDIX B: EXAMPLE CLARIFICATION OF CRITERIA

Each criterion in the 2061 procedure is clarified by a brief explanation, a set of indicators, and a scoring scheme. For example, the criterion “Providing variety of phenomena” (first criterion in Category III) is clarified as shown below:

Providing variety of phenomena. Does the material provide multiple and varied phenomena to support the benchmark idea?

Clarification. Scientists construct and use scientific knowledge to describe, explain, predict, and design real-world objects, systems, or events. For science teaching to be consistent with the nature of science, scientific ideas need to be connected to pieces of the real world. From a cognitive perspective, phenomena are important for helping make scientific ideas credible/plausible to students. This criterion examines whether the material provides a sufficient number of phenomena (defined in Webster’s New World Dictionary as “any event, circumstance, or experience that is apparent to the senses and can be scientifically described or appraised, as an eclipse”) to support the general propositions put forth in benchmarks. The material can provide experiences with phenomena directly through hands-on activities or demonstrations (firsthand experiences) or indirectly through the use of text, videos, pictures, models, etc. (Note that the next criterion “Providing vivid experiences,” examines whether experiences provided, whether firsthand or otherwise, are likely to be vivid.)

Curriculum materials need to provide experiences with phenomena whether the benchmarks they address describe phenomena or abstract ideas. If a benchmark describes a phenomenon (e.g., “water in an open container disappears”), addressing this criterion involves having

students encounter instances of this phenomenon (e.g., monitoring the water level in a fish tank). If a benchmark describes an abstract idea (e.g., “Atoms and molecules are in perpetual motion”), addressing this criterion involves presenting students with phenomena that make the idea appear credible (e.g., students sitting closer to a newly opened bottle of perfume smell it sooner than students sitting further away).

How many phenomena are necessary will depend on the difficulty of the benchmark and the level of generalization it requires. “Varied” means here that materials should provide experiences in a variety of contexts: contexts within a particular scientific field (e.g., biology), contexts from different scientific fields if applicable, and contexts from everyday life. Providing experiences in a variety of contexts is important for most benchmarks. It is particularly important for benchmarks from *Benchmarks* chap. 1 to 3 (“Nature of Science, Mathematics, and Technology”), and 11 (“Common Themes”) since understanding the ideas in these benchmarks requires encountering them and generalizing their applicability across several fields and disciplines. For example, to support the idea that “Thinking about things as systems means looking for how every part relates to others,” experiences need to be provided for a variety of systems from different scientific fields and technology, such as an ecosystem or solar system, an educational or monetary system, and/or a telephone or transportation system. But even for a relatively simple benchmark, such as “water in an open container disappears,” experiences with a variety of phenomena, such as drying out unwrapped bread, drying clothes on a clothesline, drying up of paints, monitoring the water level of an uncovered fish tank, or monitoring the level of puddles on a sidewalk, will be helpful.

Indicators of meeting the criterion

1. Phenomenon **can support** the benchmark idea.
2. Phenomenon is **explicitly linked** (in the student’s text or the teacher’s guide) to the relevant benchmark idea.

Scoring Scheme

Excellent (3): The material provides a sufficient number and variety of phenomena that meet Indicators 1 and 2 for each of the ideas examined.

Satisfactory (2): The material provides some phenomena that meet Indicators 1 and 2.

Poor (1): The material provides one phenomenon that meets Indicators 1 and 2.

None (0): No phenomena are provided that meet Indicators 1 and 2.

APPENDIX C: The Use of Van Helmont’s Experiment in *SciencePlus* and in *Food for Plants*

In the following, excerpts are provided from two curriculum materials that include van Helmont’s experiment that showed that plants do not use food from the soil to support their growth.

In *Food for Plants*, teachers are alerted that “students’ ideas about soil as food for plants are challenged as they think about an experiment done by the Belgian scientist Jan Van Helmont in 1642” (*Teacher’s Guide*, p. 39). Before they read about the experiment, students write down and justify their ideas about whether they think that soil is food for plants and make predictions about the outcome of van Helmont’s experiment. After they read about the experiment, students are encouraged to think again about whether soil is food for plants.

In *SciencePlus*, students are presented with Van Helmont's experiment (pp. 11–12), however, teachers are not alerted that this experiment could challenge students' naïve conceptions and students are not encouraged to link the experiment to their own ideas about where plants get their food.

Food for Plants (Student Book, pp. 37–40):

ACTIVITY NINE: Dr. Van Helmont

Is Soil Food for Plants?

How many people in your class think that soil might be food for plants? _____

How many people in your class think that water might be food for plants? _____

What are their reasons?

Traveling Back in Time

We have started doing experiments to find out about how plants get their food.

Scientists have been doing experiments for many, many years to find out about food for plants. Aristotle lived years ago, and he thought about plants and how they got their food to grow. In fact, people have probably been wondering about how plants make their food for thousands of years. Even before we called people "scientists", people were wondering about plants and trying to come up with explanations about how they live and grow. And scientists today are still doing experiments to find out about how plants get their food - we still do not have all the answers! Let's travel back in time 350 years. It is now the year 1642. We are in Europe. It is a time of excitement and exploration and travel in this part of the world. Some people have found what they call a New World across the ocean (it is actually a very old world to the Native Americans living in this "New" World)! And more people are getting interested in finding out about the why's of the world around us – more people are interested in and finding benefactors who will pay them to do science experiments.

We are going to meet one of these early scientists. He is a physician but he also does experiments with plants. His name is Jan Van Helmont. He is from the country of Belgium and the year 1642. He wants to visit with us today. He is going to help us think about our hypothesis that soil is food for plants. He was very interested in that hypothesis. Almost everyone back in 1642 was sure was the soil was the food for plants. Jan Van Helmont decided to prove them right (or wrong?).

Is soil food for plants?

Suppose a child was given 200 pounds of food to eat. Predict what would happen to the weight of that child as he or she gobbled up the food. Does the child's weight go up, go down, or stay the same? Write your answer under the box marked "Weight of Child":

What would happen to the weight of the food on the table as the child ate it? Does the weight of

Weight of Child	Weight of Food

the food go up, go down, or stay the same? Write your prediction under the box marked "Weight of Food".

Are minerals in the soil food for plants?

Everyone says that plants take in minerals from the soil. Minerals do not have very much weight, but they do weigh something. So do you think Dr. Van Helmont's tree took in minerals from the soil? _____

About how much weight did the tree get from the minerals? _____

Do you think this amount of minerals could explain how the tree gained 164 pounds? _____

Explain your thinking. _____

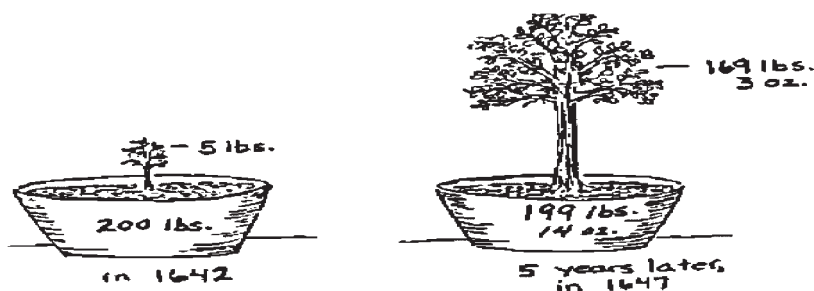
Think about Van Helmont's experiment. Does that experiment give us any evidence to say whether or not minerals in the soil are food for plants? _____

Explain your thinking. _____

Is water food for plants?

Van Helmont thought that his experiment was evidence that water must be food for plants. He thought that if soil and minerals in the soil were not giving the tree its food, then the tree must be gaining weight by getting food from the water. After all, he had been watering the tree everyday for five years.

But remember our scientific definition of food. Water helps the tree to grow, but does it give the tree energy? Could the tree live and grow if all it took in was water? Let's think about the evidence from our experiment with the grass plants. It might help us decide whether or not water is energy-giving food for plants.



Weight gain by tree	Weight loss by soil

The tree gained a whole lot of weight, but the soil did not lose hardly any weight!

What do you think Van Helmont concluded? Is soil a food for plants? Why or why not?

Van Helmont decided that soil is NOT a food for plants. The tree did not use any of the soil to grow bigger. In order to grow bigger, the tree (like all living things) needs _____ that is in food. Think about our scientific definition of food.

Does Van Helmont's experiment give us evidence to say that soil is or is not food for plants?

Explain your thinking. _____

SciencePlus (p. 11–12):**Water—How Essential Is It?**

When you started this chapter, you may have thought that plants get their food only from the soil. If you did, you are in good company. That's what the great philosopher Aristotle thought too! But a Belgian scientist of the seventeenth century, Jan Baptist van Helmont, questioned this belief and decided to look into the matter further. He decided to find out what role water had in plant growth.

What happens when a plant is deprived of water? You have probably seen what happens when you or somebody else forgets to water a houseplant. Plants that don't get enough water wilt. If they continue to be deprived of water, they eventually die.

Van Helmont performed an experiment that convinced him that water was so important that it was completely responsible for the great change in mass that occurs in growing plants. In other words, he concluded that water was completely responsible for growth in plants.

What do you think? Before you decide, let van Helmont tell you about his experiment, translated from his own words. Don't let the odd wording and spelling throw you. People spelled and spoke quite differently in van Helmont's day. (The masses have been changed to metric units.)

Journal

I have learned from this handicraft operation that all vegetables do immediately and materially proceed out of the element of water only. For I took an earthen vessel in which I put ninety kilograms of earth that had been dried in a furnace, which I moistened with rainwater, and I implanted therein the trunk, or stem, of a Willow tree, weighing two kilograms and about two-hundred and fifty grams. At length, five years being finished, the tree sprung from thence did weigh seventy-six kilograms. But I moistened the earth vessel with rain-water, or distilled water (always when there was need) ... and lest the dust should be co-mingled with the earth, I covered the lip or mouth of the vessel with an iron plate covered with tin, and easily passable with many holes. I computed not the leaves that fell off, in the four autumnes. At length, I again dried the earth of the vessel, and there were found the same ninety kilograms, wanting a few grams. Therefore almost seventy-four kilograms of wood, bark, and roots arose out of water only.

Jan Baptist van Helmont

11

ORGANIZER, continued

baking soda (sodium bicarbonate); 19 mL of water; wax pencil; lamp with 100 W bulb; materials to test for starch from Part 2 of Exploration 1 on page 7 (additional teacher materials: 1 g of sodium hydroxide and 24 mL of water for preparing sodium hydroxide solution; 50 mL of vinegar; a few strips of pH paper; 25 mL of 0.1 M sodium thiosulfate solution; a few sheets of old newspaper; 250 mL of water; 600 mL beaker; see Advance Preparation on page 1D.) **Exploration 4:** lamp with 100 W bulb; 2 test tubes; wooden

splint; a few matches; 50 g of baking soda (sodium bicarbonate); 950 mL of water; 2 clear plastic containers or 600 mL beakers; 2 clear funnels; elodea sprig; safety goggles

Teaching Resources

Exploration Worksheets, pp. 6, 8, and 9
Transparency Worksheet, p. 11
Transparency 1

★ An Exploration Worksheet is available to accompany Exploration 2 on page 10 (Teaching Resources, page 6).

Answers to**Questions, page 10**

1. The leaves from a plant left in the dark have little or no starch compared with leaves from a plant that has been in bright light.
2. Both parts suggest that green plants need light to produce starch. In each case the plants (or parts of the plants) that did not receive light had less starch. However, steps 1 and 2 could not be considered without comparing the results with those from Exploration 1.
3. Steps 3 and 4 could have been done without first doing steps 1 and 2. However, steps 1 and 2 reduce the amount of starch in the plant's leaves so that the production of starch in the uncovered parts of the leaves is more apparent.

Water—How Essential Is It?

Before students begin reading, have them discuss what happens when a plant is deprived of water. (*It wilts and eventually dies.*) Help them to conclude that water is essential for healthy growth.

Have students silently read about van Helmont's experiment, or call on a volunteer to read it aloud. Monitor students' understanding of the material by having them discuss and complete the pictorial summary on page 12.

Point out to students that before the 1600s, scientists thought that plants obtained their food directly from the soil. This theory was first proposed by the Greek philosopher Aristotle (384–322 B.C.). This food was believed to have been absorbed directly by the roots and utilized by the plants without much change. Van Helmont was the first to provide experimental evidence that conflicted with this theory.

Primary Source

Description of change: excerpted from early writings by Jan Baptist van Helmont

Rationale: excerpted to focus on the importance of water to plant growth

Answers to In-Text Questions

- A** The following data complete the summary:
- mass of willow tree at the beginning of the experiment = 2 kg and about 250 g (or about 2.25 kg)
 - mass of dried earth = 90 kg
 - mass of dried earth at the end of the experiment = 90 kg, minus a few grams
 - mass of grown tree = 76 kg
- B** Van Helmont carefully planned his experiment before he began. He took careful quantitative measurements at the beginning and end of the experiment. He was careful to prevent unknown material from entering the earthen vessel he used.
- C** Van Helmont assumed that water was the only substance that entered the tree. He also assumed that the tree absorbed all of its nutrients through its roots. Van Helmont did not realize that the tree gained a nutrient (carbon dioxide) from the air.

CROSS-DISCIPLINARY FOCUS

Mathematics

Have students repeat van Helmont's experiment using some fast-growing plants, such as bean or corn plants. Students should keep careful notes and record their data in the form of graphs. They can then share the results of their work with the class.

Homework

Have students design an imaginary experiment that refutes van Helmont's conclusion that water is solely responsible for plant growth. (One example is to try to grow a plant in the absence of carbon dioxide, such as underwater or in space.)

Here is a summary of van Helmont's experiment. Fill in the missing data in your ScienceLog. **A**

Mass of dried earth = ?



Mass of willow tree at the beginning of experiment = ?



Mass of dried earth at the end of the experiment = ?



Van Helmont's experiment indicated that water was an important raw material for plants. Somehow, plants used water to build new tissues and structures. It was left to others, however, to find out exactly how the conversion of water into plant matter takes place.

Van Helmont was a careful experimenter. Can you give some examples that show this? But even though he was **B** careful, van Helmont made a major mistake when he concluded that water alone was completely responsible for the increase in mass in plants. What do you think was wrong with his conclusion? **C**

In the next lesson you will perform Explorations involving gases that van Helmont knew nothing about. You will also be getting more information that will allow you to answer the question, "Where does the starch in plants come from?"



Mass of grown tree = ?

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